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Addressing hallucinations with RAG and NMISS in Italian healthcare LLM chatbots☆

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ABSTRACT

This work addresses hallucinations in Large Language Models (LLMs) through a knowledge-grounded evaluation framework. A Retrieval-Augmented Generation (RAG) approach is employed to ground responses in structured external data, thereby improving factual consistency in generated outputs. To refine hallucination evaluation, the Negative Missing Information Scoring System (NMISS) is introduced as a context-sensitive scoring mechanism that extends traditional overlap-based metrics by incorporating contextual evidence. Rather than uniformly penalizing non-overlapping content, NMISS distinguishes between unsupported additions, used as a proxy for hallucinations, and contextually grounded elaborations. Using Italian health news articles as a domain-specific knowledge base, the performance of various LLMs is assessed. Results show that Gemma2 and GPT-4 produce the most reference-aligned responses, while mid-tier models (e.g., Llama2, Llama3, and Mistral) benefit significantly from NMISS, demonstrating improved contextual understanding. This work contributes to more reliable hallucination evaluation by reducing the penalization of valid, context-supported information and improving the assessment of AI-assisted knowledge systems, with implications for healthcare and other structured knowledge domains, as well as for semantics-driven system design.

1. Introduction

Large Language Models (LLMs) are deep learning models typically built on the Transformer architecture, as introduced by Vaswani [1]. Designed to understand and generate human language, LLMs are trained on extensive text datasets, allowing them to grasp the nuances of language. These models perform various tasks, from basic text classification to advanced text generation, and recent iterations can also interpret images and generate code.

LLMs have surged in popularity due to their intuitive, prompt-based interfaces, enabling effortless interaction. The past few years have seen the continuous development of notable LLMs such as the GPT family of models (e.g., GPT-3.5, GPT-4), as well as systems built upon them, such as ChatGPT [2,3], alongside Gemini [4], Falcon [5], Llama [6–8], and Gemma [9,10]. These models excel in language comprehension [11,12], text summarization [13], and reasoning [14–16]. As a result, their applications span various industries, such as processing electronic medical records, matching clinical trials, and drug discovery in healthcare. In finance, LLMs assist with fraud detection and trading, while in business, they automate customer service via chatbots and virtual assistants [17].

However, LLMs face a significant challenge: hallucinations [18,19]. Hallucinations occur when models generate nonsensical or inaccurate content, failing to reflect the source material accurately [20]. This behavior is a serious threat since it affects the application of LLMs to real-world cases [21], as hallucinations can undermine the reliability of their outputs. Prior work [22]

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categorizes hallucinations into two types: factuality hallucinations, where generated content diverges from verifiable facts, and faithfulness hallucinations, where the content strays from user instructions or input context. Factuality hallucinations manifest as either factual inconsistencies or fabrications. For example, a model might mistakenly claim that "The Colosseum is located in Berlin, Germany" instead of Rome or invent details about non-existent technologies. Faithfulness hallucinations occur when the output deviates from user instructions or context. These can involve instruction inconsistencies (e.g., responding in Spanish when asked to translate into Italian), context inconsistencies (introducing unrelated information), or logical errors (e.g., miscalculating "two times four" as three).

The mitigation of hallucinations has become an urgent concern due to the widespread use of LLMs in sensitive areas such as healthcare. Addressing hallucinations is challenging due to LLMs' reliance on reinforcement learning from human feedback [23], their opaque black-box nature that hides knowledge boundaries [24,25], and their training on large datasets that may contain outdated or incorrect information [26]. Additionally, their versatility across many tasks complicates mitigation efforts, while their strong performance can make hallucinations, especially factual fabrications, harder to detect.

Researchers focused on various techniques to mitigate hallucinations. One of the most promising methods is Retrieval-Augmented Generation (RAG) [27], which incorporates external knowledge during text generation to anchor outputs in factual information. RAG addresses issues of accuracy and currency in LLM responses by using authoritative sources, generating verifiable answers, and enhancing user confidence across applications [28]. Other approaches include Prompt Engineering that involves optimizing LLMs outputs with different instructions [29,30], self-refinement through feedback and reasoning [31], where feedback on LLM outputs improves the accuracy and quality of subsequent responses, and prompt tuning, which modifies instructions during fine-tuning to improve task-specific performance [32].

Accurate evaluation of hallucinations is essential for effective mitigation. Traditional metrics such as Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [33], Bilingual Evaluation Understudy (BLEU) [34], and Metric for Evaluation of Translation with Explicit ORdering (METEOR) [35] measure lexical overlap between generated and reference text but often fail to capture semantic equivalence [36]. Recent embedding-based metrics, such as BERTScore [37], use contextual embeddings to assess semantic similarity. Another direction involves "opening the box" — analyzing token probabilities or logit outputs to detect hallucinations [19,38] — though this approach requires access to internal model states, which is not feasible for proprietary systems.

This work proposes a general, domain-independent framework for mitigating and evaluating hallucinations in LLMs. For mitigation, a RAG architecture integrates external knowledge into the generation process, ensuring that outputs remain grounded in factual information. The methodology employs a multilingual Sentence Transformer model [39,40] to construct and query a vector database for contextual retrieval. A range of LLMs, specifically Gemma [9], Gemma-2 [10], Mistral [41], GPT-4 [3], LLaMA-2 [7], and LLaMA-3 [8] are compared using a zero-shot prompting strategy that instructs models to generate answers based on retrieved context. To evaluate interpretability and task complexity, 100 questions were manually designed and curated, then reviewed by a domain expert.

For evaluation, this work introduces the **Negative Missing Information Scoring System (NMISS)**, a novel context-aware scoring refinement that extends traditional reference-based metrics by incorporating retrieved evidence. NMISS operates as a scoring adjustment mechanism, correcting overlap-based evaluation when grounded, context-supported elaborations are mistakenly penalized as lexical mismatches. Whereas existing metrics rely solely on lexical or semantic overlap with reference answers, NMISS leverages external context to assign credit to semantically valid content beyond the reference surface form. This integration of external knowledge into both generation and evaluation supports the development of knowledge-aware LLM-based systems.

In assessing hallucination mitigation through RAG using reference-based metrics, Gemma-2 and GPT-4 achieve the highest overall performance across all difficulty levels, while Gemma frequently fails to produce meaningful context-supported answers, resulting in the weakest performance. NMISS then refines this evaluation by providing a more faithful assessment of grounded yet lexically diverse responses. Substantial improvements are observed both for mid-tier systems such as LLaMA-2 and LLaMA-3, and for Gemma-2, which benefits from NMISS even at low hallucination rates due to its tendency to produce context-enriched elaborations.

The results indicate that models producing context-supported responses tend to obtain higher NMISS-adjusted scores, as their outputs frequently integrate evidence retrieved from the provided contexts. Importantly, NMISS does not reward length per se, but rather reduces evaluation bias against semantically grounded answers that diverge lexically from the reference. By emphasizing contextual faithfulness over strict surface overlap, NMISS complements traditional metrics with an evidence-aware evaluation layer.

This paper is organized as follows: Section 2 reviews the relevant literature on hallucination detection, mitigation, and evaluation. Section 3 details the methodology, highlighting the rationale of NMISS. Section 4 describes the experimental setup, and Section 5 presents the empirical findings. Finally, Section 6 summarizes contributions and outlines future research directions.

2. Related works

This section reviews recent research on hallucination management in LLMs along three main dimensions: detection techniques, which identify hallucinated content after generation; mitigation strategies, classified by resource requirements into resource-intensive and zero-resource approaches; and evaluation metrics, which assess the quality of generated responses. This organization situates the contributions of this work within the broader landscape of hallucination research.

2.1. Hallucination detection techniques

Hallucinations in LLMs have been widely studied. [22] provides a detailed overview of strategies to detect and mitigate hallucinations, offering insight into their occurrence in current LLMs architectures. Ensuring LLM's trustworthiness depends on early detection, as emphasized by [19], who highlights the importance of proactive measures during text generation.

Detection methods can be categorized into three main types [26]. First, **logit-based estimation**, which uses token-level probabilities or entropy from the model's logits [42]. Second, **verbalized confidence estimation**, where LLMs are asked to self-rate the confidence in their responses. Third, **consistency-based estimation** involves generating multiple candidate outputs and comparing them to detect logical inconsistencies, mimicking human reasoning to refine answers [43–45].

Furthermore, hallucination detection methods can be categorized into white-box, gray-box, and black-box techniques [46]. White-box methods analyze the model's internal workings to identify when incorrect statements are generated. Gray-box approaches examine logit outputs to find areas of low confidence in the output. Black-box techniques, which only require the model's final answers, are simpler to apply since they don't require access to internal states.

Recent studies introduce SAPLMA [38], which detects hallucinations by using a classifier trained to predict the truthfulness of statements based on LLMs' hidden layer activations. This approach outperforms traditional methods that rely solely on sentence probabilities. An iterative process has been proposed to detect potential hallucinations by identifying tokens with low normalized probabilities [19]. The system then retrieves relevant information to correct these errors, effectively reducing hallucinations in GPT-3.5.

2.2. Resource-intensive mitigation strategies

In addition to detection, various strategies have been developed to mitigate hallucinations in LLMs. One common approach is knowledge augmentation, such as Knowledge Injection (KI). Prior work enhances prompts with information from knowledge graphs, for instance by adding structured details such as location names for review tasks, leading to reduced hallucinations and improved factual accuracy [47]. KI-based LLMs produce more accurate assertions and can outperform larger models in response quality.

A direct approach consists in training models on structured knowledge. For example, T5 models have been trained on knowledge graph triples from Wikidata, where masking key entities enables the model to learn and generalize factual information effectively [48]. The resulting SKILL framework, after fine-tuning, outperforms standard T5 models on QA answering tasks, demonstrating the effectiveness of structured knowledge infusion.

Another key mitigation strategy is retrieval-based generation. RAG combines a sequence-to-sequence model with a dense vector index of external documents, such as Wikipedia, improving the accuracy and reliability of generated content [27]. Similar approaches have been applied in domain-specific settings, such as the biomedical field, where external sources are integrated to produce factual summaries [49].

Self-supervision has also been explored as a mitigation strategy. A recent framework introduces risk scores, referred to as POLAR scores, to estimate hallucination likelihood in LLM responses [50]. This approach uses Pareto optimization to balance supervision signals from multiple sources. The method involves collecting LLM outputs, training a probabilistic function h to estimate error rates, and optionally re-prompting the model based on the assigned risk score, potentially in combination with retrieval mechanisms. Experimental results show a significant reduction in hallucination rates across tasks such as question answering and text generation.

Finally, alignment-based approaches aim to reduce hallucinations by improving the interaction between user queries and external knowledge. For instance, the MixAlign framework aligns queries with relevant knowledge sources to resolve ambiguities, leading to improved response accuracy and a reduction in hallucinated content [51].

2.3. Zero-resource hallucination strategies

The strategies discussed in Section 2.2 are resource-intensive. In contrast, zero-resource hallucination methods are designed for black-box models and scenarios with limited resources, relying only on model outputs without needing internal access or external data. Many of these techniques not only detect hallucinations but also help mitigate them by refining responses based on internal consistency assessments.

One approach is through **self-consistency techniques**, which compare multiple responses generated by the LLM itself. [52] propose a zero-resource method that generates multiple responses from GPT-3 and compares them to a reference answer to assess factual consistency. This method detects and mitigate hallucinations using low-level metrics and outperforms traditional gray-box and black-box methods.

Another approach is **self-rating techniques**, as proposed by [53]. The authors introduce RelD, a discriminator designed to detect hallucinations in LLM-generated answers. RelD is trained by converting a regression task into a binary classification task, aligning with human intuition. By flagging low-confidence responses, it serves as an indirect mitigation mechanism. Experiments across various datasets and models (e.g., BLOOM [54]) show RelD's effectiveness, validated by both automatic metrics and human-in-the-loop evaluations. Similarly, hallucinations in fabricated references have been addressed through Direct Queries (DQs) and Indirect Queries (IQs), which operate without relying on external resources [55]. DQs consist of repeatedly asking the model binary questions to estimate a confidence probability. In contrast, IQs probe the model about the authors or content of a reference and assess consistency across multiple responses. Although primarily designed for detection, these approaches can also mitigate hallucinations

by filtering out low-confidence outputs. Experimental results show that IQs generally outperform DQs in detecting hallucinations, although performance varies across models such as GPT-4, GPT-3, and LLaMA 2.

Finally, self-contradiction has been explored as an indicator of hallucinations [56]. A prompting-based framework is introduced to detect and mitigate contradictions within the same response, relying solely on the model's internal reasoning capabilities. The framework includes steps for gathering contextual information, generating candidate statements, identifying contradictions, and refining outputs. Across multiple LLMs, this approach achieves an F1 score of approximately 80%.

2.4. Metrics for evaluating hallucinations

Traditional metrics to detect or manage hallucinations [36] are notably ROUGE [33], BLEU [34], and METEOR [35]. These indicators measure the consistency between the generated text and the source text. However, a drawback of these simple metrics is that they do not accurately capture distance word-dependencies or if the two answers diverge in lexicon but have equal meaning. BERTScore overcomes this issue [37] by leveraging pre-trained contextual embeddings from BERT [57] and matches words in candidate and reference sentences by cosine similarity, correlating well with human judgments. Recent work has introduced evaluation frameworks that explicitly assess whether generated answers are supported by retrieved evidence in retrieval-augmented generation settings. A prominent example is RAGAS [58], a reference-free evaluation framework that measures multiple aspects of RAG pipelines, including faithfulness, answer relevance, and context relevance. In particular, the Faithfulness metric evaluates whether the claims contained in a generated answer are supported by the retrieved context, assigning higher scores when a larger proportion of the response is grounded in the provided evidence. These approaches differ from traditional reference-based metrics by not relying on ground-truth answers, instead focusing on the alignment between generated outputs and retrieved evidence. However, they operate under a different evaluation paradigm and do not explicitly consider discrepancies between reference-based scores and context-supported responses in cases of lexical divergence.

2.5. Contributions

Despite significant progress in hallucination detection and mitigation, key challenges persist. In particular, traditional evaluation metrics often fail to account for contextual accuracy, unfairly penalizing semantically correct responses that differ lexically from reference answers. This paper addresses these gaps with a primary focus on evaluation. The key contributions of this work are as follows:

(1) *An experimental framework for retrieval-grounded generation*. A standard RAG-based architecture is adopted to integrate external knowledge into the generation process, ensuring that models are grounded in factual data. The framework leverages a multilingual Sentence Transformer model [39,40] for vector database generation and retrieval, along with a controlled prompting strategy. This setup enables consistent and comparable evaluation of multiple LLMs under retrieval-grounded conditions.

(2) *Comprehensive analysis of hallucinations across reasoning levels*. To evaluate hallucination tendencies in LLMs, performance is analyzed across progressively challenging levels of reasoning, categorized as low, mid, and high. This approach enables an assessment of how hallucination rates and mitigation effectiveness vary as tasks demand increasing degrees of logical complexity and understanding. The progressive evaluation highlights how model reliability and factual grounding are affected under escalating reasoning challenges.

(3) *A context-aware evaluation refinement framework*. The NMISS is introduced as a novel framework that refines traditional overlap-based metrics by incorporating contextual grounding. By analyzing residual content from machine-generated answers after comparison with human responses, NMISS identifies cases where standard metrics incorrectly penalize context-supported information, enabling a faithful evaluation of model outputs in retrieval-grounded settings.

3. Methodology

3.1. RAG architecture

RAG models combine the strengths of retrieval-based and generative language models to address issues like hallucinations and factual inaccuracies by integrating a retrieval mechanism to ground responses in relevant information [27].

Let q represent the query, c the context information retrieved from external sources, and r the output sequence generated by the model. The architecture consists of two components: (i) a retriever $p_\eta(c|q)$ with parameters η, which selects a top-K set of documents k based on the query q, and (ii) a generator $p_\theta(r_i|q, c, r_{1:i-1})$, with parameters θ, which generates each token r_i conditioned on the query q, retrieved context c, and previously generated tokens $r_{1:i-1}$.

$$p_{\text{RAG}}(r \mid q) = \sum_{c \in k(p(\cdot|q))} p_\eta(c \mid q) \prod_{i=1}^{N} p_\theta\left(r_i \mid q, c, r_{1:i-1}\right) \tag{1}$$

This RAG-sequence approach ensures consistent responses by using the same retrieved document to generate each token, maintaining coherence throughout the sequence. Here, N represents the length of the output sequence. This architecture is particularly advantageous for domains like healthcare, where factual accuracy is critical. By leveraging retrieval from medical documents, guidelines, or research articles, the RAG model ensures that responses are accurate, minimizing the risks associated with misinformation [59]. In this paper, the RAG workflow is detailed in Section 4.3.

3.2. Evaluation metrics

The metrics described in this section are used to evaluate the quality of responses generated by the system, referred to as the System Summary (SS), against the ground truth responses, referred to as the Reference Summary (RS), which are human-generated responses in this study.

Overlap-based metrics such as ROUGE, BLEU, and METEOR rely on n-gram matching, whereas BERTScore leverages contextual embeddings to measure semantic alignment beyond surface-form similarity and EM checks exact string equality.

Formally, let $S_{\text{SS}} = \{s_1, s_2, \ldots, s_l\}$ denote the multiset of n-grams in the SS, and $S_{\text{RS}} = \{r_1, r_2, \ldots, r_m\}$ denote the multiset of n-grams in the RS.

ROUGE [33] evaluates responses based on precision, recall, and F1-score. The text is pre-processed before calculating these scores [60]. Technically, ROUGE-N measures the n-gram overlap between the SS and RS. The precision, recall, and F1-score are defined as:

$$\text{Precision} = \frac{|S_{\text{SS}} \cap S_{\text{RS}}|}{|S_{\text{SS}}|} \tag{2}$$

$$\text{Recall} = \frac{|S_{\text{SS}} \cap S_{\text{RS}}|}{|S_{\text{RS}}|} \tag{3}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where $|S_{\text{SS}} \cap S_{\text{RS}}|$ denotes the number of overlapping n-grams, $|S_{\text{SS}}|$ the total number of n-grams in the SS, and $|S_{\text{RS}}|$ the total number of n-grams in the RS. Precision measures the factual accuracy of the generated text, while recall quantifies the coverage of the reference information. Accordingly, low precision suggests the presence of hallucinations, whereas low recall indicates the presence of missing or incomplete content.

Another variant used here is **ROUGE-L**, which measures the Longest Common Subsequence (LCS) between the SS and RS. The LCS represents the longest subsequence in both SS and RS while preserving word order, though the words do not need to appear consecutively [61].

BLEU [34] is designed to evaluate the quality of SS responses by comparing them against one or more RS answers. BLEU accounts for variability in word choice and word order found in different, yet valid, translations of the same source sentence. It quantifies the overlap by measuring how many n-grams from the SS are present in the RS. The core of the BLEU metric is the modified n-gram precision, which addresses the over-generation problem by treating each reference word as exhausted once a matching candidate word is found.

Let V be a function that counts the occurrences of an n-gram in a given set of n-grams. Formally, for an n-gram w and a set of n-grams S, the function V is defined as $V(w, S) = |\{s \in S \mid s = w\}|$. This function returns the number of times w appears in the set S. The modified n-gram precision p_n is calculated as:

$$p_n = \frac{\sum_{w \in S_{\text{SS}}} \min\left(V(w, S_{\text{SS}}), \max_j(V(w, S_{\text{RS}_j}))\right)}{\sum_{w \in S_{\text{SS}}} V(w, S_{\text{SS}})} \tag{5}$$

where $V(w, S_{\text{SS}})$ is the number of times the n-gram w appears in SS, and $\max_j(V(w, S_{\text{RS}_j}))$ is the maximum number of times w appears across all RS. Here, j iterates over the different reference summaries in S_{RS}, where $j = 1, 2, \ldots, m$, and m is the number of RS.

To account for translations that are too short, BLEU applies a brevity penalty (BP), defined as:

$$\text{BP} = \begin{cases} 1 & \text{if } b > r \\ e^{(1-r/b)} & \text{if } b \leq r \end{cases} \tag{6}$$

where b is the length of the candidate translation (SS), and r is the effective reference length (RS).

The final BLEU score is computed by taking the geometric mean of the modified n-gram precisions (up to length N, typically set to 4), and multiplying by the brevity penalty:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\frac{1}{N}\sum_{n=1}^{N} \log p_n\right) \tag{7}$$

METEOR [35] addresses several limitations of BLEU by emphasizing word-to-word alignment between the SS and RS, accounting for exact matches, stems, and synonyms. For each SS–RS pair, METEOR finds an alignment of unigrams that maximizes the number of matches subject to one-to-one mapping and minimal crossing. Precision (P) and recall (R) are computed as the fractions of matched unigrams in the SS and RS, respectively, and combined through a weighted harmonic mean:

$$F_{\text{mean}} = \frac{P \cdot R}{\alpha P + (1-\alpha)R},$$

where α is typically set to 0.9.

To capture word order and fragmentation, METEOR groups matched unigrams into the fewest possible number of contiguous chunks such that the words in each chunk are adjacent and in the same order in both SS and RS. Let T denote the number of such chunks, and $|V|$ the total number of matched unigrams. The fragmentation penalty is then computed as:

$$\text{Penalty} = 0.5 \left(\frac{T}{|V|} \right)^3.$$

The final METEOR score is given by:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}).$$

This formulation penalizes disordered or fragmented alignments, ensuring that longer, well-ordered matches contribute more to the final score.

BERTSCORE is an evaluation metric for text generation that leverages pre-trained BERT embeddings [37]. Unlike traditional metrics that rely on exact n-gram matches, BERTSCORE measures the semantic similarity between tokens in the SS and RS by comparing their contextual embeddings. This method captures the semantic meaning even when the exact words differ. BERTSCORE generates contextual embeddings for the tokens in both the SS and RS, where each embedding reflects the surrounding context of the word. The similarity between tokens is computed using cosine similarity. For each token in the RS, BERTSCORE identifies the most similar token in the SS, and vice versa, to compute precision and recall.

Given a RS sentence $x = \{x_1, \ldots, x_k\}$ and a SS sentence $\hat{x} = \{\hat{x}_1, \ldots, \hat{x}_l\}$, the token embeddings are represented as $\{e(x_1), \ldots, e(x_k)\}$ and $\{e(\hat{x}_1), \ldots, e(\hat{x}_l)\}$, where $e(\cdot)$ represents the contextual embedding for each token. The similarity between two embeddings is computed using cosine similarity. Precision (P_{BERT}) and recall (R_{BERT}) are calculated as follows:

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \cos(e(x_i), e(\hat{x}_j)) \tag{8}$$

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \cos(e(\hat{x}_j), e(x_i)) \tag{9}$$

The final BERTSCORE is computed as the F1 score of the precision and recall:

$$F_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \tag{10}$$

Exact Match (EM) is a straightforward metric that penalizes any response not exactly equal to the provided dataset annotation. It is analogous to conventional classification accuracy, ensuring the generated text matches the reference text precisely. The EM can be defined as:

$$EM = \begin{cases} 1 & \text{if } SS = RS \\ 0 & \text{if } SS \neq RS \end{cases} \tag{11}$$

3.3. Negative Missing Information Scoring System (NMISS)

3.3.1. Rationale

The Negative Missing Information Scoring System (NMISS)[1] is introduced as a refinement layer for reference-based overlap metrics, aiming to account for contextually supported lexical content in system outputs that would otherwise be treated as mismatches. In particular, NMISS adjusts the score produced by the underlying metric by distinguishing unsupported additions from grounded elaborations supported by the retrieved context.

The metrics discussed in Section 3.2 evaluate machine-generated responses primarily through lexical or semantic overlap with reference answers. While effective in many settings, these approaches overlook the fact that non-matching n-grams may still convey valid information when they remain consistent with the retrieved context. This limitation becomes particularly evident in retrieval-grounded generation, where a response may legitimately diverge from the reference while still being factually supported by external evidence.

In practice, LLMs often enrich their outputs with additional content. Such extensions may reflect either (i) hallucinations, introducing incorrect or irrelevant information, or (ii) contextually grounded elaborations, where the model incorporates supplementary details explicitly supported by the retrieved passage.

This distinction is illustrated by the example in Table 1, which reports an answer generated by LLaMA-3. The system expands an abstention-based reference answer by adding recommendations on hand hygiene and mask usage, as in the retrieved context. This lexical divergence is penalized by ROUGE-1 Precision (0.12), while NMISS yields a refined score of 0.56 by rewarding grounded supplementary content. Traditional metrics may therefore assign low scores even to non-hallucinated responses, misclassifying valid enrichment as lexical mismatch. NMISS addresses this limitation by assessing whether unmatched n-grams in the system output align with the external evidence, acknowledging that additional content is not inherently erroneous but may instead reflect contextual expansion that standard metrics fail to capture.

[1] Code available at: https://github.com/priolap/nmiss.

Table 1
Example of system answer

Question	How to prevent influenza?
Context	[1] [1] As outlined in the Relaunch Decree: 1. prioritize prevention to minimize impacts at all levels; 2. include the full range of health and social services in plans for managing a potential (re)emergence of an epidemic, so as to ensure continuity of care even in the event of an epidemic "reactivation" (like a "backup generator"), not only in hospitals but also at the community level; 3. ensure that investments, especially in personnel. [2] rigorous behaviors aimed at strengthening measures to contain the risk of contagion. They must internalize and adopt recommendations such as washing hands frequently and thoroughly, making extensive, correct, and appropriate use of masks, avoiding as much as possible occasions for contagion related to non-essential activities, thus avoiding unnecessary movements and behaviors associated with social and recreational relationships that create situations of spontaneous gatherings. [3] To contain the spread of infection, it is essential to respect social distancing, wear masks, and thoroughly sanitize hands. It is also crucial to self-isolate in the presence of cold-like symptoms. There are no high-risk places, only high-risk behaviors. We must convey to citizens the awareness that if the necessary precautions are observed, it is difficult to contract and transmit this infection, and almost all normal social and working activities can continue. [4] For the first time, elements of high criticality are reported regarding the spread of the virus in our country. The public is urged to avoid as much as possible events and initiatives that risk gatherings in public and private places. It is recalled that it is mandatory to adopt rigorous individual behaviors in order to limit the risk of transmission and prevent a further and more rapid worsening of the epidemic. Local transmission of the virus, widespread throughout the territory.
Reference	I'm sorry, I can't help you based on the information I have.
System	I'm sorry, I can't help you based on the information I have, as the question is not related to the Relaunch Decree but rather to influenza, which is a different type of illness than the epidemic under discussion. However, based on the context, it is noted that the Relaunch Decree emphasizes the importance of strict behaviors to minimize impacts at all levels, such as social distancing, mask use, and hand hygiene, which are generally useful measures for preventing disease transmission in general, not just the discussed epidemic.

Note: The example is translated into English; the original benchmark data are in Italian.

3.3.2. Formulation

Let $\mathcal{R} = \{r_1, r_2, \ldots, r_n\}$ denote the set of reference answers, while $\mathcal{S} = \{s_1, s_2, \ldots, s_n\}$ the set of system-generated answers, and $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ the set of retrieved context passages associated with each answer pair.

Let X denote a generic overlap-based evaluation metric introduced above (e.g., BLEU, ROUGE, or METEOR), which compares a system answer s_i against a reference answer r_i. Let $G(\cdot)$ denote the multiset of lexical n-gram units extracted from a text.

For each pair (s_i, r_i), let $\mathcal{P}_X(s_i, r_i)$ denote the multiset of lexical matching units identified between s_i and r_i according to the matching procedure induced by the metric X. The matching procedure depends on the definition of X. For purely lexical metrics such as BLEU and ROUGE, matching is based on exact n-gram overlap under the chosen preprocessing. For metrics such as METEOR, the matching may incorporate normalization steps.

The residual content of the system answer, i.e., the multiset of lexical units in s_i that are not supported by the reference under the same matching criterion, is defined via multiset difference as[2]:

$$\mathcal{N}(s_i, r_i) = G(s_i) \setminus \mathcal{P}_X(s_i, r_i),$$

Next, let $\mathcal{A}_X(s_i, c_i)$ denote the multiset of residual n-gram units in $\mathcal{N}(s_i, r_i)$ that are supported by the retrieved context passage c_i. Formally,

$$\mathcal{A}_X(s_i, c_i) = \mathcal{N}(s_i, r_i) \cap G(c_i).$$

The NMISS scoring function evaluates how well system-generated answers align with both reference answers and contextual information. For this reason, NMISS combines two components: a reference-based score and a context-based score.

The reference-based evaluation function is defined as:

$$f_{\text{ref}}(s_i, r_i) = X(s_i, r_i) \tag{12}$$

The context-based evaluation function is defined as:

$$f_{\text{cxt}}(s_i, c_i) = X\big(\mathcal{N}(s_i, r_i),\, c_i\big) \tag{13}$$

where X is computed using the same definition of metric X, but restricted to the lexical subspace.[3] This overlap approximates the set $\mathcal{A}_X(s_i, c_i)$, as it captures the portion of residual units that are also supported by the context under the matching scheme of X.

Therefore, the NMISS evaluation function is given by:

$$f_{\text{NMISS}}(s_i, r_i, c_i) = \max\left(f_{\text{ref}}(s_i, r_i), \frac{\lambda_1 \cdot f_{\text{ref}}(s_i, r_i) + \lambda_2 \cdot f_{\text{cxt}}(s_i, c_i)}{\lambda_1 + \lambda_2}\right), \tag{14}$$

[2] For sequence-based metrics such as ROUGE-L, the residual $\mathcal{N}(s_i, r_i)$ is obtained by removing the token positions participating in one LCS alignment between s_i and r_i. The cardinality $\lambda_2 = |\mathcal{A}_X(s_i, c_i)|$ is then computed as the multiset intersection of the resulting residual token sequence with $G(c_i)$. This treatment is consistent with the sequence-level nature of ROUGE-L while preserving the general NMISS framework.

[3] In the contextual component, X is applied only through its lexical overlap operator. Hence, in case of BLEU, the brevity penalty is set to 1 to fairly evaluate the context length.

where

$$\lambda_1 = \left|\mathcal{P}_X(s_i, r_i)\right|, \qquad \lambda_2 = \left|\mathcal{A}_X(s_i, c_i)\right|$$

represent the cardinalities of the reference-supported matches and context-supported residual matches, respectively. By construction, NMISS preserves the range of the underlying metric X. Therefore, if $X \in [0, 1]$, then $f_{\text{NMISS}} \in [0, 1]$.

To preserve the integrity of standard scoring metrics, NMISS introduces two key mechanisms: a weighted scheme and a maximum function. These mechanisms jointly balance reference-based and context-based contributions, mitigating potential distortions such as score inflation or deflation when contextual information is incorporated.

The weighted scheme harmonizes the relative influence of reference overlap and contextual alignment. Because contextual passages are typically much longer and information-rich than reference responses, a simple averaging would overemphasize the contextual component. To address this, the weights λ_1 and λ_2 regulate their respective contributions, ensuring that neither dominates the final score. This mechanism enables NMISS to recognize cases where a system-generated response extends beyond the reference while remaining coherent and faithful to the retrieved evidence.

The maximum function serves as a safeguard against disproportionate score drops caused by denominator inflation, in metrics where the contextual component can dilute the score. This situation often arises with verbose models that rely on contextual paraphrasing when uncertain. Although such verbosity may not introduce hallucinations, it can generate noise—information that adds limited value relative to the reference. By applying a floor, NMISS reverts to the reference-based score whenever the context-based contribution disproportionately lowers the metric due to a large λ_2 term in the denominator. This ensures rewarding genuine contextual relevance while preventing technical artifacts from suppressing the overall score.

NMISS is designed to operate independently of the retrieval component and is therefore agnostic to the nature of the retrieved context, focusing exclusively on the evaluation of generated responses given a provided context. The detailed implementation is provided in Algorithm 1.

Algorithm 1 NMISS Algorithm

Require: $\mathcal{R} = \{r_1, r_2, \ldots, r_n\}$ ▷ reference answers
Require: $\mathcal{S} = \{s_1, s_2, \ldots, s_n\}$ ▷ system answers
Require: $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ ▷ retrieved contents
Require: X ▷ base overlap-based metric
Ensure: $\mathcal{M}_{\text{NMISS}}$ ▷ multiset of NMISS scores

$\mathcal{M}_{\text{NMISS}} \leftarrow \emptyset$
for $i \leftarrow 1$ to n **do**

Stage 1: Reference-based computation
$\mathcal{P}_X(s_i, r_i) \leftarrow \textsc{MatchUnits}_X(s_i, r_i)$ ▷ metric matching units
$\mathcal{N}(s_i, r_i) \leftarrow G(s_i) \setminus \mathcal{P}_X(s_i, r_i)$ ▷ residual units
$f_{\text{ref}} \leftarrow X(s_i, r_i)$ ▷ baseline metric
$\lambda_1 \leftarrow \left|\mathcal{P}_X(s_i, r_i)\right|$

Stage 2: Context-based computation
$\mathcal{A}_X(s_i, c_i) \leftarrow \mathcal{N}(s_i, r_i) \cap G(c_i)$ ▷ context-supported residual units
$f_{\text{cxt}} \leftarrow X(\mathcal{N}(s_i, r_i),\ c_i)$ ▷ baseline metric
$\lambda_2 \leftarrow \left|\mathcal{A}_X(s_i, c_i)\right|$

Stage 3: NMISS computation
if $\lambda_1 + \lambda_2 > 0$ **then**
$f_{\text{NMISS}} \leftarrow \dfrac{\lambda_1 f_{\text{ref}} + \lambda_2 f_{\text{cxt}}}{\lambda_1 + \lambda_2}$
else
$f_{\text{NMISS}} \leftarrow f_{\text{ref}}$
end if
$f_{\text{NMISS}} \leftarrow \max\left(f_{\text{ref}},\ f_{\text{NMISS}}\right)$
$\mathcal{M}_{\text{NMISS}} \leftarrow \mathcal{M}_{\text{NMISS}} \cup \{f_{\text{NMISS}}\}$
end for

Note: $\textsc{MatchUnits}_X$ returns the multiset of lexical matching units induced by X

To formally evaluate the effectiveness of NMISS, let $\mathcal{H} = \{h_1, h_2, \ldots, h_n\}$ denote the set of hallucination flags associated with the model-generated responses $\mathcal{S} = \{s_1, s_2, \ldots, s_n\}$. Each flag $h_i \in \{0, 1\}$ indicates whether response s_i is hallucinated ($h_i = 1$) or not ($h_i = 0$).

NMISS is computed exclusively over non-hallucinated responses, as it is designed to quantify false hallucination penalties produced by overlap-based metrics when system outputs include context-supported yet verbose elaborations. Non-hallucinated labels may be obtained from gold annotations or provided by an external hallucination detection module. In this work, the annotation protocol is described in Table 2. Formally, let $\mathcal{I}$ be defined as

$$\mathcal{I} = \{1, 2, \ldots, n\}, \quad \mathcal{I}_{\text{NH}} = \{\, i \in \mathcal{I} \mid h_i = 0 \,\}$$

From this subset, extract only the *valid cases* where the traditional metric score $f_{\text{ref}}(s_i, r_i)$ lies strictly between two thresholds $\tau_{\min}$ and $\tau_{\max}$, with

$$0 \leq \tau_{\min} < \tau_{\max} \leq 1$$

This constraint filters out trivial cases corresponding to near-perfect matches or near-complete mismatches[4]:

$$\mathcal{I}_{\text{v}} = \{\, i \in \mathcal{I}_{\text{NH}} \mid \tau_{\min} < f_{\text{ref}}(s_i, r_i) < \tau_{\max} \,\}$$

The corresponding subset of responses is then defined as

$$S_{\text{v}} = \{\, s_i \mid i \in \mathcal{I}_{\text{v}} \,\}$$

where $f_{\text{ref}}(s_i, r_i)$ denotes the score computed by a traditional reference-based metric (see Stage 1 in Algorithm 1).

For each valid case $i \in \mathcal{I}_{\text{v}}$, NMISS is considered to *outperform* the traditional metric if

$$f_{\text{NMISS}}(s_i, r_i, c_i) > f_{\text{ref}}(s_i, r_i)$$

The overall percentage of NMISS outperformance across all valid cases is thus computed as

$$\frac{\left|\{\, i \in \mathcal{I}_{\text{v}} \mid f_{\text{NMISS}}(s_i, r_i, c_i) > f_{\text{ref}}(s_i, r_i) \,\}\right|}{|\mathcal{I}_{\text{v}}|} \times 100 \tag{15}$$

This formulation maintains full consistency with the operational definition of NMISS and isolates its contribution precisely where traditional metrics underperform. By focusing on ambiguous, semantically rich cases that lie outside trivial bounds, NMISS becomes a robust indicator of a model's ability to generate contextually grounded yet non-literal responses.

4. Experimental setup

4.1. Data

Data comprises text-based news in Italian, collected from various sources, and all focused on the health domain within Italy. This collection encompasses multiple thematic categories, including regional governance (25%), biomedicine (21%), media commentary (15%) and national health governance (15%). The corpus is intentionally kept heterogeneous, as it serves as the retrieval knowledge base in the RAG setting and enables evaluation across diverse healthcare-related contexts.

News dataset consists of 126,470 articles from 2010 to 2024; their frequency is shown in Table 2. A pre-processing step was employed to clean the raw text of HTML tags and other unwanted artifacts to ensure data quality. Token statistics are shown in Table 2. On average, each article contains approximately 348 tokens, with a standard deviation of 264 tokens, indicating some variability in article length. The shortest article consists of just 6 tokens, while the longest contains up to 5594 tokens. The median article length is 284 tokens, with the interquartile range spanning from 195 to 421 tokens, reflecting a diverse range of article sizes.

Table 2
Summary of token statistics for the news collection.

N. Tokens	Mean	Std	Min	25%	Median	75%	Max
44 018 696	348	264	6	195	284	421	5594

4.2. Human evaluation protocol

To build a robust evaluation framework, a dataset of 100 Italian public health questions[5] is constructed. This design intentionally focuses on the biomedical subset of the news corpus (approximately one third of the overall coverage), as this portion provides the most relevant evidence for patient-oriented health information seeking.

Questions span a range of complexity, as illustrated in Table 3. They are grouped into three difficulty levels — low, mid, and high — based on average character length, used as a coarse operational proxy for complexity[6]. These thresholds are defined to provide a consistent partition of the dataset according to increasing descriptive detail, rather than as a precise measure of semantic

[4] In this experiment, let set $\tau_{\min} = 0.01$ and $\tau_{\max} = 0.99$ to exclude degenerate cases.

[5] The full list of questions is available at https://github.com/priolap/nmiss.

[6] Respectively, low: 30 characters; mid: 40 characters; high: 50 characters.

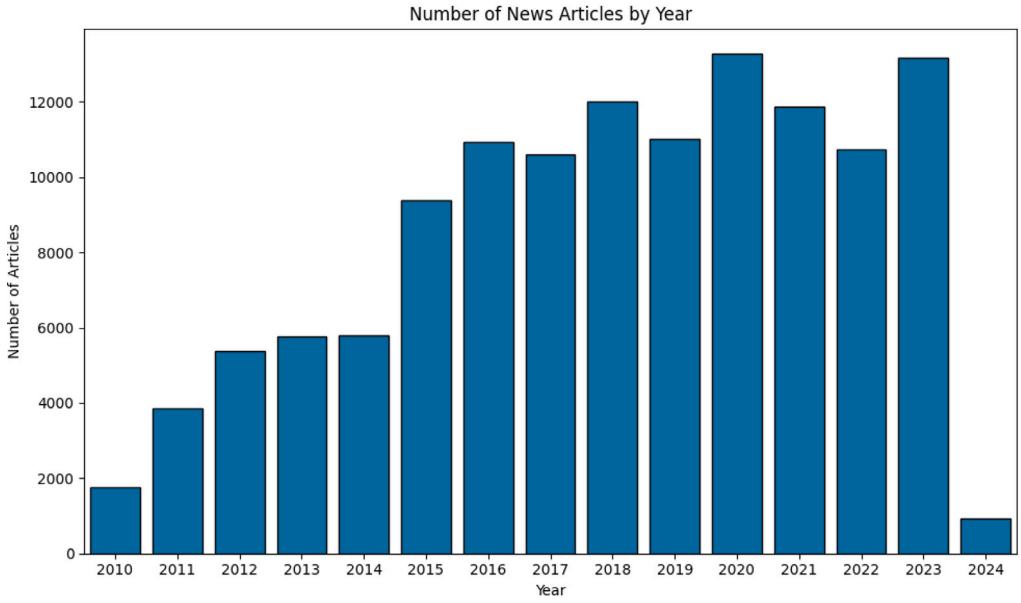


Fig. 1. News frequency aggregated by year.

Table 3
Categorized example questions.

Category	Example question
HIGH	How do new technologies help diagnose neurodegenerative diseases?
MID	What happens to the body when we are stressed?
LOW	What is a common cold?

difficulty. This choice aligns with prior work where linguistic and structural features (including length) are used as predictors of difficulty [62], although not as standalone measures.

To mitigate this limitation, an LLM-as-a-judge validation is introduced [63]. A subset is annotated with GPT-4 using criteria grounded in difficulty prediction literature (e.g., reasoning complexity and number of concepts). A soft distribution guideline is included to avoid degenerate outputs, and is used solely to calibrate the model rather than to enforce strict constraints. The prompt is reported in Table A.9.

Comparing the length-based and LLM-as-a-judge labels question by question, agreement is moderate (overall accuracy 67%) and strongly structured: the two criteria never disagree by more than one level and never confuse LOW with HIGH. Concordance is high for MID and HIGH questions (82% and 76% of items, respectively) and lower for LOW questions (44%), where the length threshold and the LLM most often diverge, the latter tending to assign borderline short questions to MID. This pattern is consistent with character length acting as a coarse but order-preserving proxy for difficulty: it captures the broad ordering of complexity while diverging from a semantic judgment precisely at the low-mid boundary (see Table 2).

Each model-generated answer was manually annotated as either *hallucinated* or *non-hallucinated* as formalized in Section 3.3.2. An answer was labeled as hallucinated if it contained factual inaccuracies, fabricated details, or information not supported by the retrieved context or the corresponding reference answer. This binary annotation dataset forms the basis for the analyses presented in Section 5, enabling direct comparison between traditional metrics and NMISS.

4.3. User-agent chatbot framework

This section outlines the architecture of the user-agent chatbot designed to handle queries related to Italian health news, using a RAG approach that integrates a Dense Vector Store (DVS) with LLMs fine-tuned for Italian language processing. This multi-model setup ensures robustness in handling diverse queries related to Italian health news, providing accurate, coherent, and contextually grounded responses.

The system begins with a Sentence Transformer model used to generate dense vector embeddings for the document database, which are then stored and indexed within the DVS. Specifically, the Paraphrase Multilingual MiniLM-L12-v2 model [39] is employed to encode both document chunks and user queries into a shared semantic vector space. The embedding step consists of transforming textual inputs into dense vector representations, while the retrieval step performs a similarity search (based on cosine similarity) over the indexed vectors to identify the top-4 most relevant document chunks. This enables efficient retrieval of relevant information

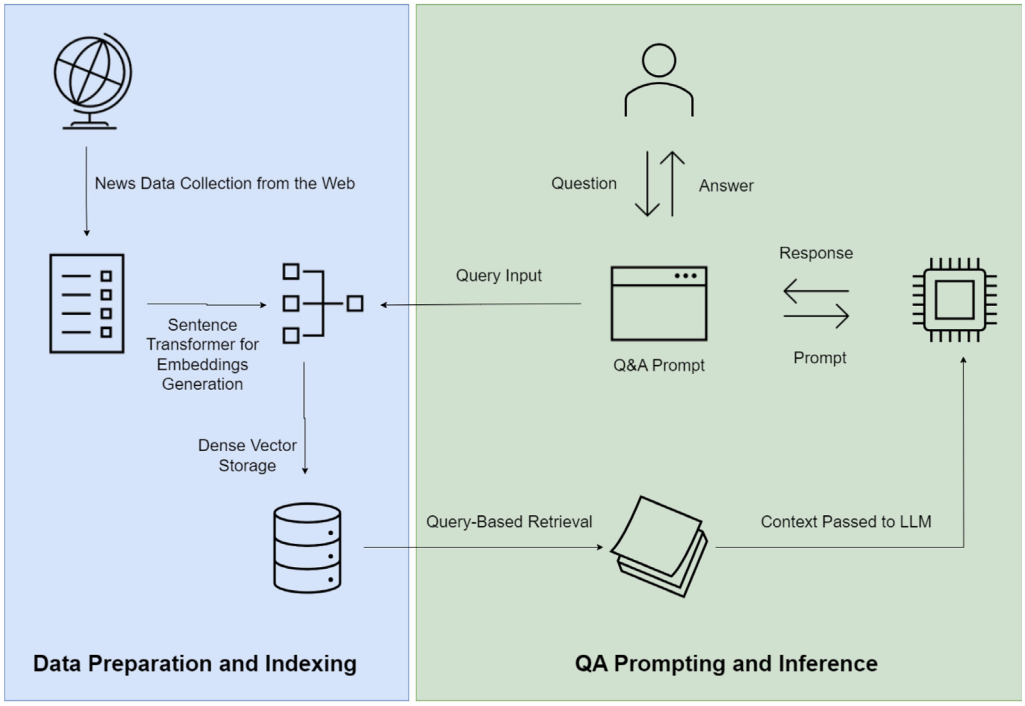


Fig. 2. User-agent application workflow.
Note. Overview of the end-to-end RAG pipeline. Text data is first split into chunks for efficient retrieval, encoded into dense vector embeddings using a Sentence Transformer model, and stored in a vector database. At inference time, the user query is embedded in the same vector space and used to retrieve the top-4 most relevant chunks via similarity search. These retrieved contexts are then provided to the LLM to generate the final response.

through semantic search. After experimenting with different chunking configurations, retrieval performance was optimized using a chunk length of 512 tokens with a 64-token overlap, evaluated on the dataset of 100 health-related queries.

For the generation component, multiple LLMs[7] capable of processing Italian text are evaluated. LLaMA-2 and LLaMA-3, developed by Meta [6,8], are optimized through Supervised Fine-Tuning (SFT) and Reinforcement Learning with Human Feedback (RLHF). Mistral [41] is a 7-billion-parameter model known for its efficiency and competitive generation accuracy. GPT-4 [3] is a multimodal model developed by OpenAI, capable of processing both textual and visual inputs. Finally, Gemma and Gemma-2 [9,10] are lightweight models from Google, fine-tuned for question-answering and summarization tasks.

The workflow illustrated in Fig. 2 illustrates the complete end-to-end RAG pipeline. The query embedding is compared against stored vectors to retrieve the top-K most semantically similar chunks from the vector database. These retrieved documents are then supplied to the LLM as contextual input. During generation, the model receives both the user query and the retrieved evidence, producing a natural language response grounded in factual data.

4.4. Prompt engineering template

A QA prompting template is developed to guide the model in generating contextually relevant answers based on the provided context. An Italian–English version is shown in Table 4. If the answer to a query is not found within the context, the chatbot responds with: "I'm sorry, I can't help you based on the information I have". This study adopts a single prompting template to ensure a controlled evaluation setting and isolate the effects of NMISS without introducing variability due to prompt engineering. Alternative prompting strategies may influence performance and are left for future work.

5. Results

5.1. RAG performance analysis

The results, summarized in Figs. 3–4, present model performance across three levels of question complexity.

[7] All open-source models were evaluated with a temperature of 0.7 for comparability.

Table 4
Prompt engineering templates (English and Italian).

System prompt (English) You are a chatbot that loves helping people. Given the following context section, answer the question using only the provided context. If the answer is not explicitly present in the context, return the following message: "I'm sorry, I can't help you based on the information I have".
System prompt (Italian) Sei un chatbot che ama aiutare le persone. Data la seguente sezione di contesto, rispondi alla domanda utilizzando esclusivamente le informazioni fornite. Se la risposta non è esplicitamente presente nel contesto, restituisci il seguente messaggio: "Mi dispiace, non so aiutarti sulla base delle informazioni che ho".
Prompt structure Context: {context} – Question: {question} – Answer: {answer}

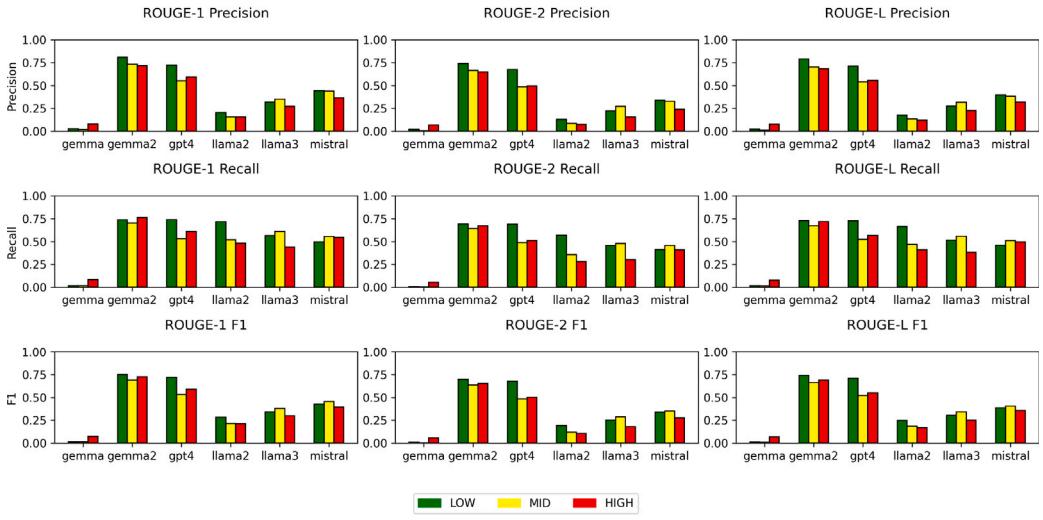


Fig. 3. ROUGE metrics by question levels.

Gemma demonstrates the weakest performance across all evaluation metrics. Its ROUGE-1 and ROUGE-2 F1 scores fall below 0.4, indicating limited ability to capture relevant unigrams and bigrams. The low ROUGE-L score further suggests poor structural coherence between generated and reference texts. BLEU and METEOR scores remain consistently low, reflecting weak lexical precision and recall, while the absence of EM confirms its difficulty in reproducing reference sequences.

In contrast, Gemma-2 achieves the strongest overall results. Its ROUGE-1 and ROUGE-2 F1 scores exceed 0.5, demonstrating strong lexical alignment, while the high ROUGE-L value indicates superior sentence-level coherence. BLEU and METEOR scores reinforce this trend, highlighting consistent precision and recall advantages.

GPT-4 follows closely behind, exhibiting balanced and reliable performance across all metrics. Although slight declines are observed at medium and high difficulty levels, it maintains high overall accuracy, achieving EM values higher than Gemma-2. These results indicate strong factual grounding and effective adherence to the given prompts.

LLaMA-2, LLaMA-3, and Mistral form a consistent mid-tier group. Their ROUGE F1 scores are generally lower, primarily due to reduced precision, though their recall remains comparable to that of the leading models. Among these, Mistral exhibits the most stable performance across all difficulty levels, whereas both LLaMA models experience sharper performance degradation as question complexity increases.

Overall, a consistent performance hierarchy is observed across all evaluation metrics: Gemma-2 and GPT-4 lead as top performers; LLaMA-2, LLaMA-3, and Mistral form a stable middle tier; and Gemma remains the weakest model, often producing responses that lack coherence or contextual grounding. This ranking pattern holds across ROUGE, BLEU, BERTScore and METEOR, indicating robustness in the comparative evaluation.

Among the individual metrics, METEOR yields consistently higher scores than BLEU for all models, as it accounts for stemming, synonymy, and paraphrase matching. This property allows to reward equivalent expressions that BLEU tends to penalize, thereby providing a more lenient yet informative measure of linguistic adequacy. A similar trend is observed for BERTScore, whose values are uniformly high across models due to its semantic sensitivity.

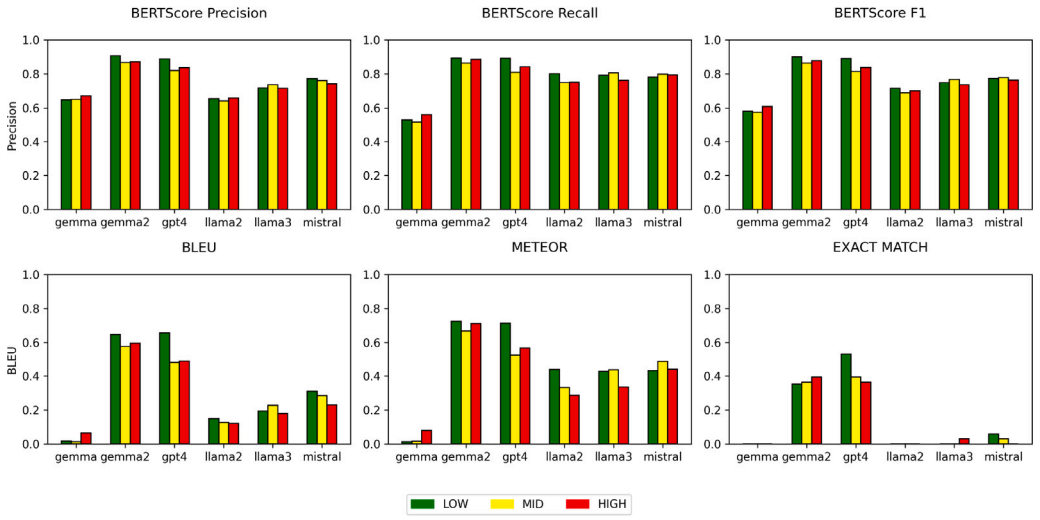


Fig. 4. BLEU, METEOR and EM metrics by question levels.

Table 5
BLEU and METEOR NMISS comparison.

Model	Level	BLEU	NMISS-BLEU	METEOR	NMISS-METEOR
gemma	LOW	0.040	0.040	0.198	0.198
gemma2	LOW	0.354	**0.645**	0.379	0.379
gpt4	LOW	0.361	**0.401**	0.575	0.575
llama2	LOW	0.191	**0.288**	0.464	0.464
llama3	LOW	0.240	**0.287**	0.532	**0.541**
mistral	LOW	0.328	**0.383**	0.500	0.500
gemma	MID	0.000	0.000	0.353	0.353
gemma2	MID	0.496	**0.738**	0.540	**0.557**
gpt4	MID	0.362	**0.377**	0.586	0.586
llama2	MID	0.172	**0.242**	0.483	**0.489**
llama3	MID	0.238	**0.311**	0.500	0.500
mistral	MID	0.381	**0.414**	0.634	**0.636**
gemma	HIGH	0.689	0.689	0.625	0.625
gemma2	HIGH	0.374	**0.727**	0.513	**0.526**
gpt4	HIGH	0.262	**0.283**	0.445	0.445
llama2	HIGH	0.193	**0.298**	0.446	0.446
llama3	HIGH	0.210	**0.326**	0.385	**0.387**
mistral	HIGH	0.335	**0.369**	0.525	0.525

Note: Average scores. Bold values highlight the highest scores per metric in NMISS. NMISS outperformance is computed over valid hallucination cases (excl. degenerate scores of 0 or 0.99).

5.2. NMISS performance evaluation

A detailed breakdown of the absolute score differences is reported in Tables 5–8. The tables show that NMISS-based metrics consistently outperform their standard counterparts across models and difficulty levels, supporting the claim that semantic grounding is not always captured by n-gram overlap alone. Tables A.10–A.12 in Appendix report the percentage of gold-annotated non-hallucinated cases in which NMISS assigns higher scores than traditional overlap-based metrics, stratified by model and question difficulty. To make the uncertainty of these percentages explicit, the outperformance values are reported with 95% Wilson confidence intervals in Table A.14. The intervals are wide whenever the number of valid cases is small, which occurs for opposite reasons across models: Gemma retains few valid cases because most of its responses are hallucinated, whereas GPT-4 retains few because its close lexical alignment with the reference places many responses above the upper validity threshold. Conversely, the precision gains for Gemma-2, LLaMA-2, and LLaMA-3 rest on more valid cases and remain clearly bounded away from zero.

Table 6
ROUGE-1 versus ROUGE-1 NMISS.

Model	Level	P	R	F1	NMISS_P	NMISS_R	NMISS_F1
gemma	LOW	0.870	0.233	0.367	0.870	0.233	0.367
gemma2	LOW	0.673	0.472	0.503	**0.778**	0.472	0.503
gpt4	LOW	0.522	0.700	0.573	**0.613**	0.700	**0.573**
llama2	LOW	0.292	0.718	0.373	**0.469**	**0.718**	**0.398**
llama3	LOW	0.438	0.713	0.453	**0.575**	0.723	**0.472**
mistral	LOW	0.543	0.584	0.505	0.632	0.584	0.507
gemma	MID	0.600	0.450	0.514	0.600	0.450	0.514
gemma2	MID	0.586	0.627	0.573	**0.866**	**0.634**	**0.624**
gpt4	MID	0.748	0.644	0.646	**0.805**	0.644	0.646
llama2	MID	0.243	0.700	0.332	**0.446**	**0.702**	**0.369**
llama3	MID	0.382	0.711	0.429	**0.569**	0.711	**0.441**
mistral	MID	0.519	0.726	0.566	**0.632**	**0.727**	**0.572**
gemma	HIGH	0.887	0.808	0.841	0.887	0.808	0.841
gemma2	HIGH	0.554	0.581	0.539	**0.812**	**0.586**	**0.574**
gpt4	HIGH	0.642	0.525	0.523	**0.720**	0.525	0.523
llama2	HIGH	0.311	0.675	0.391	**0.457**	0.675	**0.408**
llama3	HIGH	0.395	0.525	0.399	**0.583**	0.525	**0.401**
mistral	HIGH	0.474	0.629	0.490	**0.599**	0.629	**0.495**

Note: Average scores. Bold values highlight the highest scores per metric in NMISS. NMISS outperformance is computed over valid hallucination cases (excluding degenerate scores of 0 or 0.99).

Table 7
ROUGE-2 versus ROUGE-2 NMISS.

Model	Level	P	R	F1	NMISS_P	NMISS_R	NMISS_F1
gemma	LOW	0.682	0.176	0.280	0.682	0.176	0.280
gemma2	LOW	0.489	0.352	0.370	**0.724**	0.352	0.370
gpt4	LOW	0.372	0.455	0.395	**0.443**	0.455	**0.405**
llama2	LOW	0.171	0.489	0.225	**0.255**	0.489	**0.235**
llama3	LOW	0.338	0.640	0.363	**0.379**	0.640	0.363
mistral	LOW	0.421	0.525	0.419	**0.456**	0.525	**0.423**
gemma	MID	0.143	0.105	0.121	0.143	0.105	0.121
gemma2	MID	0.582	0.585	0.553	**0.860**	**0.611**	**0.640**
gpt4	MID	0.599	0.471	0.488	**0.629**	0.471	0.488
llama2	MID	0.162	0.562	0.229	**0.254**	**0.569**	**0.251**
llama3	MID	0.314	0.650	0.355	**0.398**	**0.655**	**0.372**
mistral	MID	0.422	0.647	0.477	**0.481**	**0.651**	**0.485**
gemma	HIGH	0.803	0.740	0.766	0.803	0.740	0.766
gemma2	HIGH	0.440	0.437	0.420	**0.834**	**0.468**	**0.497**
gpt4	HIGH	0.490	0.340	0.358	**0.523**	0.340	**0.361**
llama2	HIGH	0.189	0.477	0.245	**0.277**	0.477	**0.267**
llama3	HIGH	0.235	0.370	0.243	**0.419**	**0.381**	**0.274**
mistral	HIGH	0.336	0.524	0.371	**0.421**	**0.528**	**0.381**

Note: Average scores. Bold values highlight the highest scores per metric in NMISS. NMISS outperformance is computed over valid hallucination cases (excluding degenerate scores of 0 or 0.99).

Overall, NMISS yields heterogeneous improvements across models, metrics, and difficulty levels. The strongest outperformance is observed for Gemma-2, which achieves consistently high gains across all settings, particularly for BLEU (78.4% overall), indicating that NMISS better captures semantically correct answers that diverge from reference n-gram patterns.

For METEOR, NMISS improvements remain comparatively small (maximum 13.6% for Gemma-2), which is expected given METEOR's stemming and synonym-aware matching that already reduces sensitivity to lexical variation.

For ROUGE, improvements exhibit a clear component-wise structure. NMISS produces substantial gains in precision (e.g., 92.5% outperformance for Gemma-2 and above 80% for the LLaMA models, Table A.11), while gains in recall are far smaller (at most 7% in ROUGE-1 recall; see Table A.11). This asymmetry is structural, and follows directly from how the contextual sub-score f_{cxt} is normalized. For precision, f_{cxt} divides the context-supported residual by the residual length, which is short, so it is frequently high (median 0.49 for ROUGE-1) and exceeds the reference precision in 52.6% of non-hallucinated cases. For recall, the same residual is divided by the length of the full retrieved context, which is large by construction, so f_{cxt} is near zero and exceeds the reference recall in only 2.5% of cases (9 of 367 non-hallucinated cases). Since NMISS takes the maximum with respect to f_{ref} (Eq. (14)), recall can increase only in these cases, which are confined to answers with very short reference summaries and near-zero baseline recall, where a small amount of grounded residual content suffices to raise an otherwise minimal coverage score.

This component-wise behavior also shows that NMISS does not reward length per se. Because NMISS is a weighted average rather than a sum, context-supported residual content enters both the numerator (through $\lambda_2 f_{\text{cxt}}$) and the denominator (through

Table 8
ROUGE-L versus ROUGE-L NMISS scores.

ROUGE-L versus ROUGE-L NMISS							
Model	Level	P	R	F1	NMISS_P	NMISS_R	NMISS_F1
gemma	LOW	0.696	0.186	0.294	0.696	0.186	0.294
gemma-2	LOW	0.630	0.447	0.475	**0.763**	0.447	0.475
gpt4	LOW	0.474	0.619	0.515	**0.542**	0.619	**0.515**
LLaMA-2	LOW	0.229	0.609	0.298	**0.354**	0.609	**0.308**
LLaMA-3	LOW	0.385	0.661	0.404	**0.472**	**0.669**	**0.419**
mistral	LOW	0.462	0.523	0.441	**0.544**	0.523	**0.447**
gemma	MID	0.333	0.250	0.286	0.333	0.250	0.286
gemma-2	MID	0.554	0.582	0.537	**0.866**	**0.588**	**0.607**
gpt4	MID	0.701	0.594	0.600	**0.734**	0.594	0.600
LLaMA-2	MID	0.209	0.633	0.288	**0.339**	**0.636**	**0.308**
LLaMA-3	MID	0.346	0.659	0.389	**0.500**	0.659	**0.401**
mistral	MID	0.465	0.675	0.517	**0.580**	**0.678**	**0.528**
gemma	HIGH	0.887	0.808	0.841	0.887	0.808	0.841
gemma-2	HIGH	0.488	0.500	0.471	**0.800**	**0.511**	**0.520**
gpt4	HIGH	0.570	0.439	0.447	**0.630**	0.439	0.447
LLaMA-2	HIGH	0.243	0.562	0.310	**0.361**	0.562	**0.326**
LLaMA-3	HIGH	0.314	0.438	0.319	**0.513**	**0.440**	**0.334**
mistral	HIGH	0.411	0.573	0.434	**0.528**	**0.574**	**0.440**

Note: Average scores. Bold values highlight the highest scores per metric in NMISS. NMISS outperformance is computed over valid hallucination cases (excl. degenerate scores of 0 or 0.99).

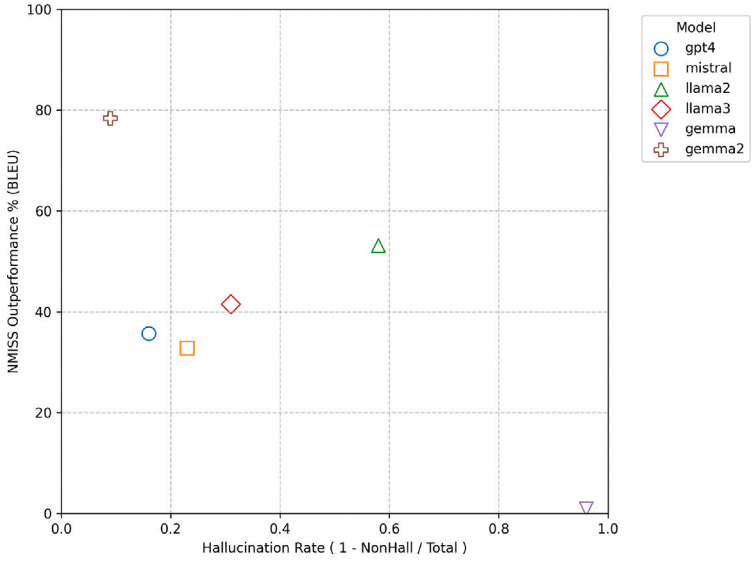


Fig. 5. NMISS utility vs. Hallucination rate (BLEU).

λ_2): adding grounded content draws the score towards f_{cxt} while increasing the weight that normalizes it, so the result stays bounded between f_{ref} and f_{cxt} and cannot grow without bound with output length. This is the purpose of the weighting scheme introduced in Section 3.3.2. Consistent with this design, a length-driven effect would raise recall together with precision, whereas in practice the gains concentrate almost entirely in precision (Tables A.11–A.12), confirming that NMISS adds credit only where overlap metrics penalize grounded content, without inflating reference coverage. The relationship between these gains and verbatim overlap with the retrieved context is examined separately in Section 5.3.

As a result, ROUGE F1 gains are moderate and model-dependent, with LLaMA-2 showing the most stable improvements overall (26.2% and 27.5% for ROUGE-1 and ROUGE-2 F1). This behavior is consistent with NMISS acting as a scoring-layer refinement, particularly in the RAG setting, where correct responses may rely on context-supported terms that are not explicitly present in the reference. In contrast, GPT-4 exhibits only marginal NMISS-ROUGE gains, consistent with its closer lexical alignment with reference answers and strong EM performance (Section 5.1).

Finally, Mistral shows small but consistent gains across metrics, while Gemma yields no measurable NMISS advantage, indicating minimal divergence from overlap-based scoring in this setting.

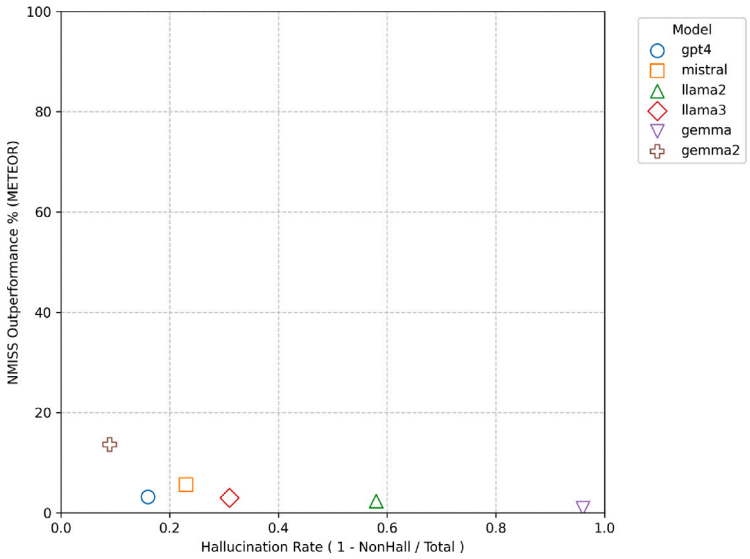


Fig. 6. NMISS utility vs. Hallucination rate (METEOR).

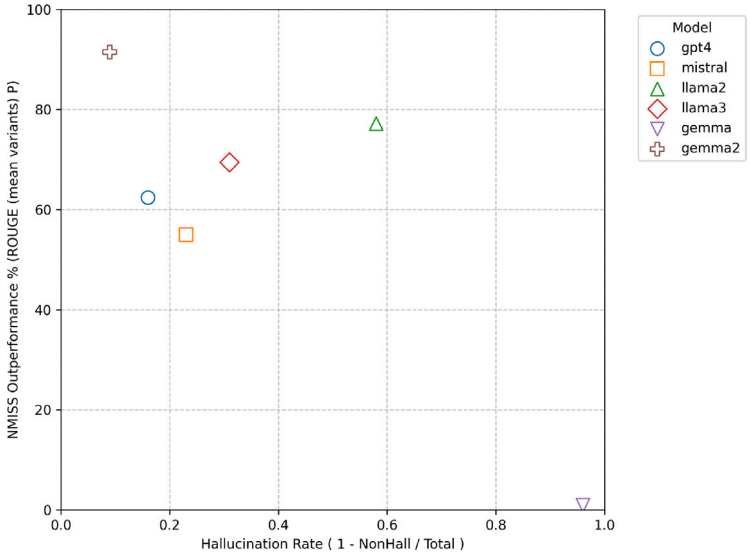


Fig. 7. NMISS utility vs. Hallucination rate (ROUGE Precision).

Figs. 5–8 further explore the relationship between model-level hallucination rates and NMISS outperformance. For BLEU (Fig. 5), the largest improvement is observed for Gemma-2, which combines a low hallucination rate with strong NMISS gains. This suggests that even highly grounded models may be underestimated by strict overlap metrics when they produce more elaborated answers.

At the same time, LLaMA-2 and LLaMA-3 exhibit robust NMISS improvements despite higher hallucination rates, indicating that part of the apparent performance gap in mid-tier systems may stem from surface-form divergence rather than purely factual errors. Overall, NMISS helps disentangle lexical mismatch from genuine hallucination effects, providing a more faithful assessment of grounded response quality.

In the case of METEOR (Fig. 6), the effect is less pronounced. Since METEOR already incorporates stemming, synonymy, and paraphrase matching, it captures a substantial portion of the semantic variability between system outputs and references. As a result, the set of unmatched lexical items is relatively small, limiting the impact of NMISS's contextual component. Under these conditions, NMISS mainly refines borderline cases rather than introducing large score shifts. Nonetheless, Gemma-2 and Mistral still exhibit meaningful gains, indicating that their elaborations remain grounded even when lexical overlap is already high.

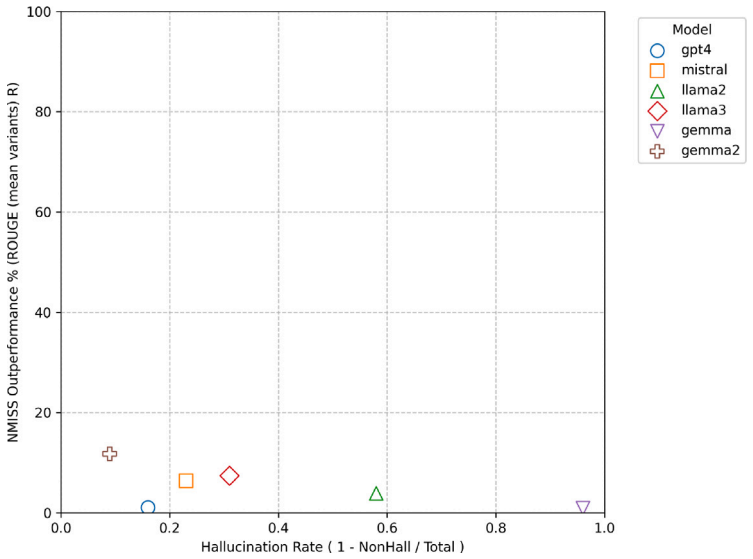


Fig. 8. NMISS utility vs. Hallucination rate (ROUGE Recall).

ROUGE analyses (Figs. 7–8) further examine how NMISS outperformance relates to model-level hallucination rates. For ROUGE precision, the strongest NMISS gains are observed for models that produce lexically diverse yet grounded outputs, suggesting that overlap-based scoring may underestimate their quality even at relatively low hallucination rates. ROUGE recall exhibits weaker and less systematic associations with hallucination tendency, indicating that NMISS utility in this component is not primarily driven by coverage effects.

Overall, NMISS proves most informative in two complementary scenarios: (i) for mid-tier models producing grounded but lexically variable answers, where overlap-based metrics may underestimate correctness due to surface-form mismatch; and (ii) for high-performing systems such as Gemma-2, where responses remain highly accurate yet include additional context-supported elaborations beyond the reference phrasing. These results highlight NMISS as a robust complement to traditional lexical overlap evaluation across models and difficulty levels.

5.3. Limitations

Despite its advantages, this framework has limitations that warrant discussion. First, evaluation is conducted on manually curated questions in the Italian healthcare domain. While this controlled setting enables fine-grained analysis of hallucination behavior, it may limit generalizability. Although NMISS is language- and domain-agnostic — operating on structural relations between outputs, references, and retrieved context — validation across domains and languages is left to future work.

Second, NMISS extends overlap-based metrics and does not incorporate embedding- or LLM-based approaches (e.g., BERTScore). This design favors interpretability and controllability but may miss aspects of semantic similarity. Integrating semantic representations constitutes a natural direction for future work.

Third, the evaluation relies on gold hallucination labels, requiring manual annotation and limiting scalability. While consistent with prior work and ensuring high-quality supervision, this step could be replaced by including an automatic hallucination detector in downstream applications.

Finally, NMISS may reward responses that reproduce retrieved context verbatim. To substantiate this concern empirically, the per-instance NMISS gain on non-hallucinated valid cases is related to a copy-rate, defined as the proportion of system tokens also present in the retrieved context (Table A.13). The sensitivity to copying is real but strongly metric-dependent: it is pronounced for BLEU, where high-overlap answers contribute most of the gain, moderate for ROUGE-1 precision, and negligible for METEOR. Gains nonetheless persist across the full copy-rate spectrum, including low-overlap answers, indicating that NMISS credits genuinely divergent grounded content rather than copying alone. Although its weighting scheme mitigates score inflation, relevant copied content can still be favored without deeper reasoning. This issue is partly inherent to overlap-based metrics and is inherited by NMISS rather than introduced by it; it is most pronounced for a pure-precision metric such as BLEU and least for METEOR. Moreover, NMISS is applied only to non-hallucinated responses, and in a retrieval-grounded setting where the model is instructed to answer from the provided context, a high overlap reflects expected grounding rather than a defect. This highlights the need for future extensions that better distinguish grounded reasoning from surface-level reuse.

6. Conclusions

This study investigates hallucination behavior in LLMs from a dual perspective: mitigation through a RAG architecture and evaluation refinement via the proposed NMISS metric. Experiments are conducted on a question-answering task over Italian health news articles, simulating user-agent interactions in a sensitive knowledge domain. Six models — Gemma, Gemma-2, GPT-4, LLaMA-2, LLaMA-3, and Mistral — are evaluated in a zero-shot setting under a fixed prompting template.

To assess hallucination mitigation through retrieval grounding, this work first evaluates RAG-generated outputs using traditional reference-based metrics, complemented by BERTScore as a semantic similarity measure beyond surface lexical overlap. Gemma-2 and GPT-4 achieve the strongest overall performance. Gemma-2 exhibits consistently high precision and recall across difficulty levels, while GPT-4 attains the highest EM accuracy, reflecting a closer alignment with reference phrasing. Mid-tier models (LLaMA-2, LLaMA-3, and Mistral) generate contextually relevant yet lexically diverse responses, which are often underestimated by strict overlap-based evaluation. This tendency is reflected in their ROUGE component structure, where recall is often higher than precision, indicating that they capture much of the reference content while diverging in surface form.

This recall–precision asymmetry directly motivates NMISS, which introduces a context-aware scoring refinement that assigns credit to grounded content beyond the reference surface form. Applied to gold non-hallucinated responses, NMISS consistently recovers semantic value overlooked by overlap-based metrics. In particular, mid-tier models such as LLaMA-2, LLaMA-3, and Mistral exhibit substantial gains, indicating that their grounded yet lexically diverse elaborations are often underestimated by strict overlap scoring. Gemma-2 also benefits from NMISS despite low hallucination rates, reflecting its tendency to produce context-enriched answers, whereas GPT-4 shows smaller improvements, consistent with its closer lexical alignment to reference responses.

Taken together, these findings suggest that RAG architectures mitigate hallucinations by grounding model outputs, while NMISS complements this process by providing a faithful evaluation of contextual correctness and informativeness. This combined approach is particularly relevant in high-stakes domains such as healthcare, where factual grounding outweighs surface-level similarity.

Future work may extend NMISS beyond to embedding-based and LLM-based evaluation frameworks, as well as to multilingual and multi-hop question answering settings. Additionally, its integration into unified pipelines combining retrieval-based mitigation, automatic hallucination detection, and context-aware evaluation objectives remains an open direction. More robust strategies for question difficulty classification, such as topic-aware criteria, will also be explored.

Beyond the Italian healthcare domain, the proposed RAG–NMISS framework is applicable to other languages and knowledge areas where reliable, evidence-based generation is essential. Broadly, the framework is relevant to Semantics-driven Systems Engineering, as it supports evaluating LLM outputs against both reference answers and retrieved knowledge. In this sense, NMISS contributes to validating knowledge-aware system behavior while remaining compatible with modular pipelines that treat retrieval, hallucination detection, and evaluation as distinct components. Although the present study adopts a standard RAG setup, the approach is not tied to a specific knowledge source and may also extend to graph-based retrieval settings, including hybrid LLM–KG architectures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A.9
Prompt used for LLM-as-a-judge level classification.

Instruction	Classify the difficulty of the following question as LOW, MID or HIGH.
Criteria	Evaluation considers linguistic and cognitive factors: reasoning complexity, number of concepts, syntactic structure, ambiguity/negation, and required domain knowledge.
Guidelines	LOW: single concept, simple structure; MID: multiple concepts or moderate reasoning; HIGH: multi-step reasoning, complex interpretation, or advanced knowledge.
Distribution	Expected distribution: LOW ∼30%, MID ∼40%, HIGH ∼30%.
Output	Respond with one label only: LOW, MID, or HIGH.
Settings	Model: GPT4, temperature = 0.

Table A.10
NMISS outperformance across BLEU-METEOR and question levels.

Metric	Model	Low	Mid	High	Overall
BLEU	gemma	0.0	0.0	0.0	0.0
BLEU	gemma-2	**87.5**	**75.0**	**76.5**	**78.4**
BLEU	gpt4	66.7	28.6	26.7	35.7
BLEU	LLaMA-2	55.6	41.7	63.6	53.1
BLEU	LLaMA-3	33.3	45.0	46.7	41.5
BLEU	mistral	33.3	33.3	31.6	32.7
METEOR	gemma	0.0	0.0	0.0	0.0
METEOR	gemma-2	0.0	**21.4**	**15.0**	**13.6**
METEOR	gpt4	0.0	14.3	0.0	3.2
METEOR	LLaMA-2	0.0	6.3	0.0	2.4
METEOR	LLaMA-3	**4.8**	0.0	4.3	3.0
METEOR	mistral	0.0	11.1	3.7	5.6

Values indicate the percentage of valid non-hallucination cases in which NMISS outperforms the corresponding traditional metric. Bold denotes the best score per metric. ‘‘Overall’’ averages across difficulty levels.

Table A.11
NMISS Outperformance across ROUGE (P, R) variants and question levels.

Metric	Model	Low	Mid	High	Overall
ROUGE-1 P	gemma	0.0	0.0	0.0	0.0
ROUGE-1 P	gemma-2	90.0	**92.3**	**94.1**	**92.5**
ROUGE-1 P	gpt4	71.4	60.0	69.2	68.0
ROUGE-1 P	LLaMA-2	**91.7**	87.5	78.6	85.7
ROUGE-1 P	LLaMA-3	81.0	85.7	82.6	83.1
ROUGE-1 P	mistral	52.9	70.4	76.9	68.6
ROUGE-2 P	gemma	0.0	0.0	0.0	0.0
ROUGE-2 P	gemma-2	**90.0**	**83.3**	**93.8**	**89.5**
ROUGE-2 P	gpt4	71.4	16.7	53.8	50.0
ROUGE-2 P	LLaMA-2	66.7	46.7	76.9	62.5
ROUGE-2 P	LLaMA-3	30.0	57.9	61.9	50.0
ROUGE-2 P	mistral	20.0	34.6	52.0	37.9
ROUGE-L P	gemma	0.0	0.0	0.0	0.0
ROUGE-L P	gemma-2	**90.0**	**92.3**	**94.1**	**92.5**
ROUGE-L P	gpt4	71.4	50.0	76.9	69.2
ROUGE-L P	LLaMA-2	91.7	81.3	78.6	83.3
ROUGE-L P	LLaMA-3	76.2	71.4	78.3	75.4
ROUGE-L P	mistral	58.8	66.7	50.0	58.6
ROUGE-1 R	gemma	0.0	0.0	0.0	0.0
ROUGE-1 R	gemma-2	0.0	**15.4**	**5.0**	**7.0**
ROUGE-1 R	gpt4	0.0	0.0	0.0	0.0
ROUGE-1 R	LLaMA-2	**0.0**	8.3	0.0	2.9
ROUGE-1 R	LLaMA-3	9.1	0.0	0.0	2.3
ROUGE-1 R	mistral	0.0	6.7	0.0	2.3
ROUGE-2 R	gemma	0.0	0.0	0.0	0.0
ROUGE-2 R	gemma-2	0.0	**25.0**	**21.1**	**16.7**
ROUGE-2 R	gpt4	0.0	0.0	5.9	3.2
ROUGE-2 R	LLaMA-2	**0.0**	18.2	0.0	6.1
ROUGE-2 R	LLaMA-3	10.0	8.3	23.5	15.4
ROUGE-2 R	mistral	0.0	7.1	17.6	10.0
ROUGE-L R	gemma	0.0	0.0	0.0	0.0
ROUGE-L R	gemma-2	0.0	**15.4**	**15.0**	**11.6**
ROUGE-L R	gpt4	0.0	0.0	0.0	0.0
ROUGE-L R	LLaMA-2	**0.0**	8.3	0.0	2.9
ROUGE-L R	LLaMA-3	9.1	0.0	5.3	4.7
ROUGE-L R	mistral	0.0	13.3	5.9	7.0

Note. Values indicate the percentage of valid non-hallucination cases in which NMISS outperforms the corresponding traditional metric. Bold denotes the best score per metric. ‘‘Overall’’ is computed over all valid cases pooled across difficulty levels, and therefore need not equal the simple mean of the three level-specific values.

Table A.12
NMISS outperformance across ROUGE F1 variants and question levels.

Metric	Model	Low	Mid	High	Overall
ROUGE-1 F1	gemma	0.0	0.0	0.0	0.0
ROUGE-1 F1	gemma-2	0.0	**40.0**	**20.0**	**21.7**
ROUGE-1 F1	gpt4	0.0	0.0	5.9	3.2
ROUGE-1 F1	LLaMA-2	**25.0**	31.3	21.4	26.2
ROUGE-1 F1	LLaMA-3	9.5	13.0	8.7	10.4
ROUGE-1 F1	mistral	5.9	3.7	11.1	7.0
ROUGE-2 F1	gemma	0.0	0.0	0.0	0.0
ROUGE-2 F1	gemma-2	0.0	**30.8**	**31.6**	**23.3**
ROUGE-2 F1	gpt4	14.3	0.0	5.9	6.5
ROUGE-2 F1	LLaMA-2	**25.0**	26.7	30.8	27.5
ROUGE-2 F1	LLaMA-3	0.0	14.3	33.3	16.1
ROUGE-2 F1	mistral	6.7	7.7	11.5	9.0
ROUGE-L F1	gemma	0.0	0.0	0.0	0.0
ROUGE-L F1	gemma-2	0.0	**46.7**	**30.0**	**28.3**
ROUGE-L F1	gpt4	0.0	0.0	0.0	0.0
ROUGE-L F1	LLaMA-2	**16.7**	25.0	14.3	19.0
ROUGE-L F1	LLaMA-3	4.8	17.4	17.4	13.4
ROUGE-L F1	mistral	5.9	7.4	11.1	8.5

Note. Values indicate the percentage of valid non-hallucination cases in which NMISS outperforms the corresponding traditional metric. Bold denotes the best score per metric. "Overall" is computed over all valid cases pooled across difficulty levels, and therefore need not equal the simple mean of the three level-specific values.

Table A.13
Sensitivity of NMISS gains to verbatim context overlap.

Metric	Spearman ρ (copy-rate, gain)	% cases copy-rate ≥ 0.8	% gain from copy-rate ≥ 0.8
BLEU	0.58***	41.3	78.6
ROUGE-1 P	0.32***	41.1	52.3
METEOR	0.08	41.5	48.9

Note. Computed over non-hallucinated valid cases ($0.01 < f_{\text{ref}} < 0.99$). Copy-rate is the proportion of system tokens also present in the retrieved context. "% gain" is the share of the total positive NMISS gain contributed by answers with copy-rate ≥ 0.8.
The set of valid cases is metric-dependent, hence the small differences in the "% cases" column. For METEOR the share of gain from high-overlap answers (48.9%) tracks their share of cases (41.5%), consistent with the absence of a copy-driven effect.
*** Denotes $p < 0.001$.
The METEOR correlation is not significant ($p = 0.18$).

Table A.14
NMISS outperformance by difficulty level, with 95% Wilson CIs.

Metric	Model	LOW	MID	HIGH	Overall
BLEU	gemma2	88 [53, 98] (n = 8)	75 [47, 91] (n = 12)	76 [53, 90] (n = 17)	78 [63, 89] (n = 37)
	gpt4	67 [30, 90] (n = 6)	29 [8, 64] (n = 7)	27 [11, 52] (n = 15)	36 [21, 54] (n = 28)
	llama2	56 [27, 81] (n = 9)	42 [19, 68] (n = 12)	64 [35, 85] (n = 11)	53 [36, 69] (n = 32)
	llama3	33 [16, 56] (n = 18)	45 [26, 66] (n = 20)	47 [25, 70] (n = 15)	42 [29, 55] (n = 53)
	mistral	33 [14, 61] (n = 12)	33 [18, 53] (n = 24)	32 [15, 54] (n = 19)	33 [22, 46] (n = 55)
	gemma	0 [0, 79] (n = 1)	–	0 [0, 66] (n = 2)	0 [0, 56] (n = 3)
ROUGE-1P	gemma2	90 [60, 98] (n = 10)	92 [67, 99] (n = 13)	94 [73, 99] (n = 17)	92 [80, 97] (n = 40)
	gpt4	71 [36, 92] (n = 7)	60 [23, 88] (n = 5)	69 [42, 87] (n = 13)	68 [48, 83] (n = 25)
	llama2	92 [65, 98] (n = 12)	88 [64, 96] (n = 16)	79 [52, 92] (n = 14)	86 [72, 93] (n = 42)
	llama3	81 [60, 92] (n = 21)	86 [65, 95] (n = 21)	83 [63, 93] (n = 23)	83 [72, 90] (n = 65)
	mistral	53 [31, 74] (n = 17)	70 [52, 84] (n = 27)	77 [58, 89] (n = 26)	69 [57, 78] (n = 70)
	gemma	0 [0, 79] (n = 1)	0 [0, 79] (n = 1)	0 [0, 66] (n = 2)	0 [0, 49] (n = 4)
ROUGE-1F1	gemma2	0 [0, 26] (n = 11)	40 [20, 64] (n = 15)	20 [8, 42] (n = 20)	22 [12, 36] (n = 46)
	gpt4	0 [0, 35] (n = 7)	0 [0, 35] (n = 7)	6 [1, 27] (n = 17)	3 [1, 16] (n = 31)
	llama2	25 [9, 53] (n = 12)	31 [14, 56] (n = 16)	21 [8, 48] (n = 14)	26 [15, 41] (n = 42)
	llama3	10 [3, 29] (n = 21)	13 [4, 32] (n = 23)	9 [2, 27] (n = 23)	10 [5, 20] (n = 67)
	mistral	6 [1, 27] (n = 17)	4 [1, 18] (n = 27)	11 [4, 28] (n = 27)	7 [3, 15] (n = 71)
	gemma	0 [0, 79] (n = 1)	0 [0, 79] (n = 1)	0 [0, 66] (n = 2)	0 [0, 49] (n = 4)

(*continued on next page*)

Table A.14 (*continued*).

Metric	Model	LOW	MID	HIGH	Overall
METEOR	gemma2	0 [0, 28] (n = 10)	21 [8, 48] (n = 14)	15 [5, 36] (n = 20)	14 [6, 27] (n = 44)
	gpt4	0 [0, 35] (n = 7)	14 [3, 51] (n = 7)	0 [0, 18] (n = 17)	3 [1, 16] (n = 31)
	llama2	0 [0, 24] (n = 12)	6 [1, 28] (n = 16)	0 [0, 22] (n = 14)	2 [0, 12] (n = 42)
	llama3	5 [1, 23] (n = 21)	0 [0, 14] (n = 23)	4 [1, 21] (n = 23)	3 [1, 10] (n = 67)
	mistral	0 [0, 18] (n = 17)	11 [4, 28] (n = 27)	4 [1, 18] (n = 27)	6 [2, 14] (n = 71)
	gemma	0 [0, 79] (n = 1)	0 [0, 79] (n = 1)	0 [0, 79] (n = 1)	0 [0, 56] (n = 3)

Note. Per-cell NMISS outperformance (%) over valid non-hallucinated cases ($0.01 < f_{\text{ref}} < 0.99$), with 95% Wilson confidence intervals [64] and the number of valid cases n.

Data availability

The authors do not have permission to share data.

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